# Jucheng Shen

jucheng.shen@rice.edu | juchengshen.github.io | github.com/juchengshen

## Education

**Rice University** Expected May 2027

*B.A. in Computer Science, Mathematics, and Mathematical Economic Analysis, Major GPA – 3.8/4.0*

- Relevant Coursework: Reinforcement Learning, Computer Vision, Honors Linear Algebra, Honors ODE, Real Analysis, Differential Geometry, Abstract Algebra, Advanced Algorithms and Data Structures

## Research Experience

**California Institute of Technology** May 2026 – Present

*Research Intern · Advisor: Prof. Anima Anandkumar*

- Investigate how the spectral structure of continuous language representations affects learning efficiency and generalization in diffusion language models.
- Lead the project, develop spectral regularization and attenuation methods, and train and evaluate models on language modeling, length generalization, translation, and summarization.

**Rice University** Jan 2026 – Sep 2026

*Research Intern · Advisor: Prof. Anastasios Kyrillidis*

- Studied how asymmetric input injection enables a shared-weight recurrent Transformer to develop distinct computational roles for reasoning.
- Proposed *Asymmetric Input Recurrence* (*AIR*) and designed ablation and representation analyses, showing that a clear state-identity signal can induce functional specialization and improve performance on selected abstract reasoning tasks.

**Princeton University** Nov 2025 – Feb 2026

*Research Intern · Advisor: Prof. Zhuang Liu*

- Developed *i1*, a fully open text-to-image model trained from scratch, with systematic exploration of data, architecture, and training choices.
- Built the dataset selection and filtering pipeline and ran 300+ ablation experiments to evaluate modeling and training decisions.

**UT Austin & Intel Labs** May 2025 – Sep 2025

*Research Intern · Advisors: Prof. Atlas Wang & Dr. Souvik Kundu*

- Developed training-free adaptive decoding methods that improve diffusion language model throughput while maintaining generation quality.
- Proposed *OSDT* and led the development of *CadLLM*, designing adaptive decoding strategies and conducting experiments that demonstrated throughput gains of 24–50% and up to 2.3×, respectively.

**Rice University** Apr 2025 – Aug 2025

*Research Intern · Advisor: Prof. Yuke Wang*

- Developed *SuperGen*, a system for generating 2–4K videos through low-resolution generation, upscaling, and tiled refinement without fitting the full-resolution video on a single GPU.
- Implemented *SuperGen* on two baselines, conducted end-to-end evaluations and ablation studies, and contributed figures and manuscript text.

## Publications

[1] **One Model, Two Roles: Emergent Specialization in a Shared Recurrent Transformer**
**Jucheng Shen***, Barbara Su*, Anastasios Kyrillidis
*NeurIPS 2026*

[2] **Improving the Throughput of Diffusion-based Large Language Models via a Training-Free Confidence-Aware Calibration**
**Jucheng Shen**, Gaurav Sarkar, Yeonju Ro, Sharath Nittur Sridhar, Zhangyang Wang, Aditya Akella, Souvik Kundu
*Findings of ACL 2026*

[3] **Beyond Static Cutoffs: One-Shot Dynamic Thresholding for Diffusion Language Models**
**Jucheng Shen**, Yeonju Ro
*NeurIPS 2025 Efficient Reasoning Workshop*

[4] **i1: A Simple and Fully Open Recipe for Strong Text-to-Image Models**
Boya Zeng, Tianze Luo, Shu Pu, **Jucheng Shen**, Taiming Lu, Gabriel Sarch, Zhuang Liu
*NeurIPS 2026 BeNTo Workshop*

[5] **HyperGen: Empowering Ultra-high-resolution Video Generation Framework with Sketch-Tile Collaboration**
Fanjiang Ye, Zepeng Zhao, Yi Mu, **Jucheng Shen**, Renjie Li, Kaijian Wang, Saurabh Agarwal, Myungjin Lee, Triston Cao, Aditya Akella, Arvind Krishnamurthy, T. S. Eugene Ng, Zhengzhong Tu, Yuke Wang
*PPoPP 2027*

## Teaching Experience

**Rice University** Jan 2026 – Present
*Teaching Assistant · COMP 282: Computational Optimization of AI*

## Academic Service

**Reviewer**

- NeurIPS 2025 Efficient Reasoning Workshop
- COLM 2026 Efficient Reasoning Workshop

## Leadership Experience

**HackRice** Apr 2025 – Oct 2025
*Board Member* (*Tracks and Workshops*)

- Co-designed and shipped the official starter code used by 500+ hackers, and designed competition tracks and judging criteria for HackRice 15.
- Hosted 2 technical workshops for 100+ attendees; ran Q&A and live coding demos.

**Rice Chinese Theater Club** Nov 2023 – Nov 2024
*President*

- Led a team of 30+ members in organizing semester productions, coordinating venues, licensing, sets and props, publicity, and budgeting.
- Performed in three different plays with a combined audience of over 1,000 across all performances, and represented the club at quarterly meetings of the North American College Chinese Theater Alliance.

## Honors & Awards

Frank Liu Jr. Prize for Creative Innovations in Music, Fashion, and the Arts in Napier Rice Launch Challenge 2023.